Artificial Intelligence Solution Implementation in Fraud Detection

1. Business context

Financial fraud is one of the biggest challenges in digital payments, significantly impacting the global economy, with a cumulative impact valued at approximately GBP 3.89 trillion over the past 20 years (Ahmad, Ciupac-Ulici, and Beju, 2021). The European Parliament and Council (2017, cited in Ahmad, Ciupac-Ulici and Beju, 2021, p. 2) defines financial fraud as 'an act of deception committed intentionally for personal interests or to produce a loss to another party'.

Artificial Intelligence (AI) plays a key role in protecting against financial fraud. Thanks to its adaptive nature, it can identify patterns often missed by human analysts and has been proven effective in several financial fraud detection areas (Sarna *et al.*, 2025). However, the use of AI in fraud detection also presents challenges, particularly in handling class imbalance and balancing the trade-off between detecting fraudulent transactions and minimising false positives (Sarna *et al.*, 2025). If not addressed correctly, these challenges could lead to suboptimal outcomes (Heaton, 2017). Additionally, there are ethical and regulatory challenges, including avoiding bias and ensuring appropriate data anonymisation (Sarna *et al.*, 2025). While these considerations are important, they are beyond the scope of this paper.

Before ML was commonly used in this space, fraud detection primarily relied on rule-based systems (RBS), where expert analysts defined rules based on historical fraud patterns (Khanum, Chaitra, and Singh, 2024). While effective in static environments, RBSs struggle to adapt to rapidly changing fraud strategies in digital payments. ML is therefore considered a key complementing tool to the traditional RBS, enabling a more dynamic, data-driven detection that improves performance (Khanum, Chaitra, and Singh, 2024; Sarna *et al.*, 2025).

This paper presents an evaluation of supervised machine learning (ML) for transactional fraud detection, focusing on model selection and model performance evaluation for imbalanced fraud detection data. The evaluation follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology (Gómez, 2021), covering business understanding, data understanding and preparation, and modeling.

2. Data selection

For this experiment, the IEEE-CIS Fraud Detection dataset from Kaggle (2019) was selected, which contains hundreds of thousands of transaction records with both numerical and categorical features, and approximately 400 variables. This includes labelled instances of fraudulent and legitimate transactions, making it well-suited for supervised fraud detection tasks. Its scale, high dimensionality, and heterogeneity reflect the complexity of real-world financial data, and its origin as part of a large-scale competition further supports its use as a benchmark dataset (Khanum, Chaitra, and Singh, 2024).

The dataset also presents several challenges. A large proportion of features are anonymised and engineered, limiting interpretability and restricting domain-level understanding. Additionally, significant missing data increases preprocessing complexity and the risk of noisy or redundant features. These represent real-world fraud detection scenarios, where data quality and interpretability are often constrained. At the same time, this dataset provides a suitable basis for evaluating the robustness of ML models for fraud detection and the effectiveness of feature engineering approaches.

3. Feature Engineering

Feature engineering is an iterative cycle that helps us inform our modeling decisions and iteratively improve them (Duboue, 2020). The cycle follows this process iteratively

Exploratory data analysis (EDA), baseline analysis, modeling, error analysis, feature engineering, and evaluation.

Following the CRISP-DM methodology (Gómez, 2021), EDA was conducted to understand the structure and quality of the dataset. To ensure computational efficiency in this experiment, the dataset was limited to 100,000 instances. Analysis revealed 394 features, including 376 floating-point variables, 4 integer variables, and 14 categorical features, confirming a highly heterogeneous feature space. Data quality was assessed through analysis of missing values, duplicates, and feature variance. Given the anonymised nature of the dataset, statistical methods were used to guide data cleaning

decisions, in line with Duboue (2020). This ensured feature representations are suitable for downstream modeling. Eight features were identified with more than 80% missing values and were removed due to their limited contribution to model learning. No duplicate records were found, suggesting that the dataset does not contain redundant instances. Two features with zero variance were identified and removed, as they provide no discriminative value. Categorical data was encoded using one-hot encoding to prepare for more efficient modeling.

The target variable '*isFraud*' was identified and separated from input features. Analysis of the class distribution revealed a highly imbalanced dataset, with 97.4% class 0 (legitimate transactions) and 2.6% class 1 (fraudulent transactions). One key implication of this imbalance is the choice of evaluation metrics, which focus on the balance between false positives (FP) and false negatives (FN), including F1-score, Recall, MCC and PR-AUC (Duboue, 2020).

Furthermore, several features exhibit high skewness. However, through statistics visualisation, we learn some have very limited variation and nearly constant values, therefore not behaving as continuous variables (e.g., V305). As Duboue (2020) highlights, transformations such as logarithmic scaling are not appropriate for these features (Duboue, 2020). In contrast, features with large value ranges and extreme skewness (e.g., V319) were transformed using logarithmic scaling to reduce the influence of outliers and improve representation. Checking for correlation with the target variable shows that no features exhibit strong linear relationships with the target class.

4. Model selection

Given the heterogeneous, high-dimensional, and highly imbalanced nature of the dataset, and the observation that no single feature is strongly predictive, it is hypothesised that effective prediction requires combining multiple features and capturing non-linear relationships. This hypothesis aligns with domain-knowledge in the fraud detection space, where the combination of multiple features, such as the time, geographic location, and amount of the transaction together, is often used to signal an anomaly (Khanum, Chaitra, and Singh, 2024). Russell and Norvig (2022) highlight that ensemble tree-based models are well-suited to this setting, as they can model feature interactions while remaining robust in complex, noisy environments.

To further explore how different learning strategies perform in this context, two ensemble approaches were selected: Random Forest (RF) and Gradient Boosting (GB). RF builds trees independently and aggregates their predictions, improving stability and reducing variance, while GB constructs trees sequentially, with each model focusing on correcting the errors of the previous one (Russell and Norvig, 2022). This comparison provides insight into how parallel versus sequential learning affects performance in imbalanced fraud detection tasks.

5. Experiment setup and results

5.1. Baseline model: Firstly, a baseline model was implemented using the DummyClassifier available in the ‘sklearn’ Python library. This provides a benchmark against which more complex models can be evaluated (Witten, 2016). As seen in Figure 1, the DummyClassifier achieved high accuracy while failing to detect any fraudulent transactions, as evident in the evaluation results and the received error. This highlights the limitations of accuracy as a metric in imbalanced datasets, reinforcing the need to evaluate models using metrics that reflect the balance between FP and FN. Additionally, it serves as an important reference point to demonstrate the effectiveness of the selected ML models.

```
              precision    recall  f1-score   support

           0       0.97      1.00      0.99     19488
           1       0.00      0.00      0.00       512

    accuracy                           0.97     20000
   macro avg       0.49      0.50      0.49     20000
weighted avg       0.95      0.97      0.96     20000
```

```
/opt/anaconda3/lib/python3.13/site-packages/sklearn/metrics/_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and
being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
/opt/anaconda3/lib/python3.13/site-packages/sklearn/metrics/_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and
being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
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being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
```

Figure 1 (*Source: Author’s own via a Jupyter Notebook, 2026*)

5.2. RF Model: An RF classifier was trained and evaluated. As seen in Table 1, it significantly surpassed the baseline model, with precision at 97%, recall at 42%, and F1-score at 59%. As expected, the PR-AUC of 0.666 far exceeded the baseline result of 0.025. The MCC of 0.635 further reflects strong performance, providing a balanced measure across both classes. As Duboue (2022) recommends, a confusion matrix—a type of contingency table—was constructed with results consistent with the analysed metrics.

As shown in Table 2, an error analysis based on this confusion matrix indicates that the RF model struggles with false negatives, missing a substantial number of fraudulent transactions. This suggests that, despite strong overall performance, detecting minority class instances remains challenging.

Confusion matrix - Random forest		
	0	1
0	19481	7
1	295	217

Table 2 (*Source: Author's own via a Jupyter Notebook, 2026*)

Using hyperparameter tuning techniques, such as adjusting tree depth, the number of trees, and the number of features considered at each split (Witten, 2016), can improve the detection of fraudulent transactions by reducing false negatives. As shown in Table 1 and Table 3, tuning the RF model increased recall for the fraud class, but also led to a substantial increase in FP. This reflects the inherent trade-off between precision and recall commonly encountered in real-world fraud detection. Similar patterns were observed across multiple tested thresholds, reinforcing the consistency of this trade-off.

Confusion matrix - Tuned Random forest		
	0	1
0	17744	1744
1	159	353

Table 3 (*Source: Author's own via a Jupyter Notebook, 2026*)

5.3. GB model: Similarly to RF, the GB model significantly outperformed the baseline model. However, as seen in Table 1, it lags behind the RF model across all tested metrics. Overall, the model exhibits a more conservative behaviour compared to RF, prioritising precision over recall.

Additionally, GB does not support missing values natively, unlike RF, and therefore requires median imputation prior to training. While this introduces an additional preprocessing step, applying the same imputation to RF showed no negative impact on performance.

5.4. Comparing Models: Both models significantly outperform the baseline classifier, which predicts only the majority class and fails to detect any fraudulent transactions. While it achieves high accuracy due to class imbalance, it provides no practical value in fraud detection. In contrast, both RF and GB demonstrate the ability to identify fraudulent transactions, confirming the hypothesis of ensemble, tree-based models successfully being applied to this problem.

Both RF with GB's learning approaches demonstrate similar overall accuracy and a strong bias toward correctly classifying legitimate transactions. However, a fundamental difference in learning strategies is evident through each model's ability to identify fraudulent transactions. RF achieves a higher recall, identifying a larger proportion of fraudulent transactions. In contrast, GB achieves a similar precision but a much lower recall, making it more conservative in flagging fraud.

Metrics/Models	**Dummy**	**RF**	**GB**	**RF Tuned**
Percision				
Class 0	97%	99%	98%	99%
Class 1	N/A	97%	91%	17%
Recall				
Class 0	100%	100%	100%	92%
Class 1	N/A	42%	32%	69%
F1-Score				
Class 0	99%	99%	99%	95%
Class 1	N/A	59%	47%	27%
MCC	/	0.635	0.532	0.303
PR-AUC	0.025	0.666	0.561	0.5

Table 1 *(source: Author's own via a Jupyter Notebook, 2026)*

To further support the hypothesis that ensemble tree-based models are well suited to this problem, 5-fold stratified cross-validation was applied (Duboue, 2022) to the better-performing model—RF. The results showed consistent performance across folds, with PR-AUC values ranging between 0.646 and 0.670 and a mean of 0.658. This is closely aligned with the original train-test split result of 0.666, indicating that the observed performance is not random or dependent on a specific data split. These findings reinforce the suitability of ensemble tree-based approaches for fraud detection in high-dimensional, imbalanced datasets.

7. Learnings for real-world application

This study demonstrates the feasibility of applying supervised, tree-based ensemble ML techniques to transactional fraud detection using a structured CRISP-DM approach. The results show that both RF and GB significantly outperform the baseline model, confirming the value of data-driven methods in identifying fraudulent activity. However, the evaluation also highlights the inherent trade-off between precision and recall, with models either prioritising fraud detection or minimising false positives. This trade-off is particularly important in real-world financial systems, where the cost of missed fraud must be balanced against the impact of incorrectly flagged legitimate transactions. The findings suggest that model selection and configuration should be guided not only by predictive performance but also by business objectives and operational constraints. Furthermore, the approach and methods demonstrated in this experiment can be transferred to real-world applications, supporting organisations in developing scalable and adaptive fraud detection systems. Lastly, the results also highlight the importance of iterative feature engineering and evaluation, as emphasised by Duboue (2020).

There are additional critical challenges inherent to fraud detection that were not addressed in this experiment but must be considered in real-world applications. For example, digital transaction systems require low-latency predictions and efficient computational performance due to the high volume and speed of transactions. As a result, models must be evaluated not only in terms of predictive performance but also computational efficiency before deployment (Khanum, Chaitra, and Singh, 2024).

Furthermore, fraud detection systems operate within strict regulatory frameworks. Regulations such as the General Data Protection Regulation (GDPR) impose requirements on data privacy, while emerging frameworks such as the EU AI Act (2024) emphasise transparency, fairness, and accountability in AI systems. These regulations require a degree of interpretability in model decisions and mandate regular auditing to ensure that deployed systems remain compliant, unbiased, and safe. Lastly, ethical considerations must also be taken into account to ensure that fraud detection systems do not introduce or reinforce bias against specific groups. This represents a key responsibility for AI practitioners beyond regulatory compliance (Leslie, 2019; Sarna et al., 2025). While these aspects are beyond the scope of this paper, they are essential for the responsible deployment of AI in real-world fraud detection systems.

Appendix

Confusion matrix - Random forest		
	0	1
0	19481	7
1	295	217

Table 2 *(Source: Author's own via Python on a Jupyter Notebook, 2026)*

Confusion matrix - Gradient Boosting		
	0	1
0	19471	17
1	348	164

Table 3 *(Source: Author's own via Python on a Jupyter Notebook, 2026)*

Confusion matrix - Tuned Random forest		
	0	1
0	17923	1565
1	181	331

Table 4 *(Source: Author's own via Python on a Jupyter Notebook, 2026)*

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